Selection Is the Bottleneck: A Diagnostic for Training-Free Referring Image Segmentation and a Training-Free Repair

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***Abstract*—Training-free referring image segmentation (RIS) is universally implemented as propose-then-select: a frozen class-agnostic segmenter emits a mask pool, a frozen vision–language model scores each candidate against the expression, and the highest-scoring candidate is returned. Work in the area concentrates on the proposal pool, on feature fusion, or on adding larger external models, on the assumption that the pool is what limits accuracy. We test that assumption directly by decomposing the *oracle gap* — the IoU available in the pool minus the IoU the selector returns — into attributable error classes, on the full RefCOCO, RefCOCO+ and RefCOCOg validation splits and at two operating points of the released HybridGL pipeline. The decomposition is decisive: *wrong-instance* selection, in which the returned mask is a different object of the right category, carries 77–86 % of all recoverable IoU on all three benchmarks and at both operating points tested, while under-segmentation — the class that mask-granularity remedies address — carries 5–23 %. We then show *why* the released guidance cannot repair this: its answer is structurally confined to the three highest-scoring proposals under the full expression, and on wrong-instance failures the referent lies outside that set 69.8 % of the time (mean score-rank 9.1), while it lies inside the top-8 of a *category* score — the same scorer reading the expression with its spatial vocabulary removed — 70.1 % of the time. Acting on that finding, we give a training-free operator that enlarges the candidate set to the category score and applies the pipeline's own coherence score subject to the expression's spatial relation. On full RefCOCO val it raises the released pipeline from 0.4482 to 0.5029 mIoU and from 0.3816 to 0.4431 oIoU ($\Delta = +0.0546$ mIoU, 95 % CI $[+0.0493, +0.0601]$, paired bootstrap $p < 0.0001$; $+0.1510$ over the unguided argmax), with no training, no ground truth and no additional forward pass. We report the operator's limits as carefully as its gains: its mIoU improvement is statistically indistinguishable from a one-line containment snap, it is neutral on RefCOCOg and RefCOCO+, and doubling the proposal density makes the released pipeline *worse* while leaving the diagnosis intact.**

***Index Terms*—Referring image segmentation, zero-shot segmentation, training-free methods, CLIP, SAM, model diagnosis.**

I. Introduction

Referring image segmentation (RIS) takes an image and a free-form referring expression and returns the mask of the described object. In the training-free or zero-shot regime, where no pixel-level supervision and no task-specific fine-tuning are used, essentially every published method follows the same template: a frozen class-agnostic segmenter — almost always SAM [1] — emits a pool of candidate masks, a frozen vision–language model — almost always CLIP [2] — scores each candidate against the expression, and the highest-scoring candidate is returned. Differences between methods lie in how the score is formed [3]–[5], in which additional cues are introduced [6], [7], or in whether an external multimodal model is consulted [8].

The organising assumption behind this body of work is that the *proposal pool* is the limiting factor, and that progress consists of obtaining better masks, better features, or better auxiliary supervision. The assumption is rarely tested. When it is, the test is usually an oracle utilisation figure — the ratio of achieved IoU to the best IoU available in the pool — which is reported as evidence that the segmenter has room to improve, without asking which part of the discarded IoU was discarded *by the selector* and why. GLOD-RIS [5], for instance, reports an oracle utilisation of roughly 62–65 % oIoU and concludes that the ceiling is mask recall; an oracle utilisation of 62 % equally means that 38 % of the IoU that *was* available was thrown away at selection time.

This paper asks the question in a form that can be settled by measurement: *when a training-free RIS pipeline fails, is the failure in the pool or in the selector, and if it is in the selector, what kind of selection error is it and is it repairable without training?*

We answer it by decomposing the oracle gap — the difference between the IoU of the best pool member and the IoU of the returned mask — into attributable classes, using only quantities that the pipeline already computes. The decomposition (§III) yields one dominant class. *Wrong instance*: the selector returns a mask that is very nearly disjoint from the referent but belongs to the correct category. It carries 82.2 % of the recoverable IoU at the unguided operating point and 77.4 % at the released HybridGL operating point on RefCOCO val, and 86.4 % / 80.1 % at the released point on RefCOCOg / RefCOCO+ respectively. Under-segmentation — the granularity class that mask-merging and mask-refinement remedies target — carries 14.3 % and 17.6 % at the two full-split RefCOCO operating points and never exceeds 22.6 % in any configuration we measured. Wrong-instance is overwhelmingly a *spatial reference* failure: 84 % of those expressions contain a spatial word, against 67 % of the split overall when probed with the same broad lexical list.

The mechanism follows from the diagnosis. The released HybridGL pipeline does contain spatial machinery — a relationship-guidance term and a GEM attention × position-mask coherence term — but it applies both only inside

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$\mathrm{topk}(\mathrm{score}, 3)$ under the full expression, so its answer is always one of the three highest-scoring proposals. On the wrong-instance failures the referent is inside that set only 30.2 % of the time (mean score-rank 9.1). Yet a perfect chooser restricted to the released top-3 would already reach 0.5812 mIoU against the 0.4482 achieved, and the referent does reach the top-8 of a *category* score — the same scorer reading the expression with its spatial vocabulary removed — for 70.1 % of those failures.

That is an actionable structural statement, and §IV acts on it with a training-free operator that changes only the candidate set and the final selection step: score proposals with the category score, take its top k, and if the expression carries a spatial relation, return the member the relation selects, ranked by the pipeline's own coherence score. The operator adds no training, no ground truth, and no forward pass.

Our contributions are:

- **A decomposition of the oracle gap in training-free RIS** (§III), showing that wrong-instance selection rather than mask quality carries 77–86 % of all recoverable IoU, measured on the full validation splits of RefCOCO, RefCOCO+ and RefCOCOg, at two operating points on RefCOCO and one on RefCOCO+/g, and surviving a doubling of proposal density.
- **A mechanism-level account of why the released guidance cannot repair it** (§III), in terms of a measurable confinement of the pipeline's answer to a top-3 candidate set, together with the ceiling that confinement imposes.
- **A training-free operator that does repair a substantial part of it** (§IV, §V), raising full RefCOCO val from 0.4482 to 0.5029 mIoU and 0.3816 to 0.4431 oIoU over the released pipeline, with the gain concentrated exactly where the diagnosis predicts.
- **A negative-result set and a measured boundary** (§V, §VI): the operator's mIoU gain is statistically indistinguishable from a one-line containment snap; it is neutral on RefCOCOg and RefCOCO+, where spatial expressions are 3–36× rarer; an MLLM chooser baseline was not run and is disclosed as the largest open flank.

We deliberately report the boundary of the repair rather than tuning it away. The diagnostic is the durable contribution; the operator is what the diagnostic buys on the benchmark where spatial reference is common, and RefCOCO+ — collected with location words excluded — is used as a *pre-registered* test of that reading.

II. Related Work

A. Training-free referring image segmentation

Zero-shot RIS was established by Global-Local CLIP [3], which scores mask proposals with CLIP and fuses local (masked-region) and global (context) representations. Ref-Diff [6] uses diffusion attention with SAM; TAS [7] adds BLIP-2 captions and negative prompts; LGD [8] obtains attribute and surround descriptions from LLaVA; GLOD-RIS [5] adds orthogonal decoupling of the foreground and background CLIP directions and a grid localization prompt to an external multimodal model. HybridGL [4], our baseline, fuses global features into the local branch (G2L) and adds three guidance modules — spatial relationship, GEM attention coherence, and a position prior.

Two properties are common to all of these. First, the final answer is an argmax over one candidate mask. Second, the weights that combine sub-scores are global constants rather than functions of the candidate pool being scored. No published method in this family calibrates its score against the other candidates of the image it is scoring. Our diagnosis is about precisely that stage, and our operator is the first, to our knowledge, to report a pool-level decomposition of where the discarded IoU goes.

B. Diagnostic and oracle analyses

Oracle and ceiling analyses are standard in proposal-based detection [9], [10] but have not been applied to training-free RIS. The word *oracle* does not appear in the recent multimodal referring segmentation survey, nor in the GRES/gRefCOCO [11] or GREx [12] benchmarks, which instead document that no training-free method has been evaluated on their generalized setting. What the field reports is oracle utilisation — achieved IoU over best-available IoU — which aggregates pool quality and selection quality into one number and therefore cannot say which is binding. We separate the two, and find the selector binding.

C. Spatial and relational reasoning in RIS

Spatial language is a long-standing sub-problem of referring expression comprehension and segmentation: modular spatial relations [13], speaker-listener-reinforcer architectures [14], and, in the training-free regime, keyword-based proposal pruning in ProposalCLIP [15] and the relationship module of HybridGL itself. ProposalCLIP is the closest prior art on the text side: it also removes or exploits locative words when scoring proposals, but it prunes with hand-specified colour, position and direction heuristics on top of a class-name-driven score and does not diagnose selection failure. HybridGL's relationship module is closer to ours structurally, and our experiments treat it as the baseline rather than as an alternative: our operator acts on top of the released pipeline at its own operating point, and we report the decomposition of the gain between candidate-set enlargement and the relation gate (§V-E).

D. Mask merging and granularity remedies

A separate family targets SAM over- and under-segmentation by merging masks: E-SAM [16] merges adjacent similar SAM masks, GF-SAM [17] groups masks by connectivity under coverage gating, and ProposalCLIP merges proposals by IoU and CLIP similarity. MARS [18] gates proposals within a tolerance of the maximum score and unions them. These target the granularity class that our decomposition measures at 2–18 % of the recoverable IoU. We include one such operator — a containment snap — as a required baseline, and find that it and our selector are complementary rather than competing.

E. Selection over proposal sets

Selecting an arbitrary subset of frozen proposals has been studied as a learned problem (RAS [19] trains a selection head on roughly two million mask groups with a fine-tuned LLaVA). We work in the training-free regime by construction: the operator we report adds no parameters and no forward pass, and its inputs are quantities the baseline pipeline already computes.

III. Diagnosis: Where the Recoverable IoU Goes

A. Setting and notation

A training-free RIS pipeline is a pair $(\mathcal{P}, S)$. The proposer $\mathcal{P}$ maps an image x to a pool of N candidate masks $M = \{m_1, \ldots, m_N\}$. The scorer S maps a mask, the image and an expression e to a scalar. The pipeline returns $m_{\hat{\imath}}$ with $\hat{\imath} = \arg\max_i S(m_i, x, e)$; the released HybridGL pipeline instead returns an argmax over a guidance-weighted score restricted to a small candidate set, which §III-D makes precise.

For a ground-truth mask g we define the *oracle* proposal $\imath^\star = \arg\max_i \mathrm{IoU}(m_i, g)$ and the *oracle gap*

$$\Gamma \;=\; \mathrm{IoU}(m_{\imath^\star}, g) \;-\; \mathrm{IoU}(m_{\hat{\imath}}, g).$$

Γ is a property of (x, g, M) and the selector only: the oracle itself does not depend on S. The oracle IoU measures what the pool makes available; Γ measures what the selector discards.

B. An error taxonomy of the oracle gap

We classify each expression by the geometric relation between the returned mask and the oracle mask, using only mask geometry and the ground truth:

- **correct** — the returned mask already has $\mathrm{IoU} \geq 0.5$ with g;
- **under-segmentation** — the returned mask is a strict subset of the oracle (the oracle is a strict superset and covers $\geq 95\,\%$ of it);
- **over-segmentation** — the oracle is a strict subset of the returned mask;
- **wrong-instance** — the two masks are near-disjoint, $\mathrm{IoU} < 0.1$, and neither contains the other;
- **boundary** — overlapping, neither containing the other, and not near-disjoint.

The classes are disjoint and exhaustive by construction. Containment is evaluated before the disjointness test, so a tiny mask inside a large one is never miscounted as a separate instance.

Table I and Fig. 1 give the result. At the unguided operating point the selector discards 82.2 % of the recoverable IoU on wrong-instance failures; at the released operating point, where the guidance modules are active, 77.4 %. Under-segmentation — the class addressed by mask merging, mask refinement, and containment-based remedies — accounts for 14.3 % and 17.6 % respectively. Over-segmentation is negligible at every operating point ($\leq$ 1.9 %), which is itself informative: the selector essentially never errs by returning a mask that contains the referent.

TABLE I

Error taxonomy of the oracle gap. Each cell is *prevalence / share of total gap mass*. Full RefCOCO validation split, 10,834 expressions, oracle mIoU 0.8221 in both columns. Under-seg. = under-segmentation; over-seg. = over-segmentation.

	unguided argmax (mIoU 0.3518)	**released guidance** (mIoU 0.4482)
correct	0.371 / 0.022	0.478 / 0.027
under-seg.	0.109 / 0.143	0.108 / 0.176
over-seg.	0.011 / 0.010	0.016 / 0.019
wrong-instance	**0.493 / 0.822**	**0.380 / 0.774**
boundary	0.015 / 0.003	0.018 / 0.004

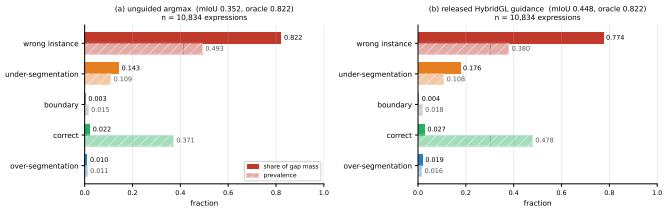


Fig. 1. Oracle-gap taxonomy on full RefCOCO val, at the unguided argmax operating point (left) and at the released HybridGL operating point (right). Solid bars: share of the total recoverable IoU. Hatched bars: prevalence. Wrong-instance selection dominates the gap mass at both operating points, although the guidance modules reduce its prevalence; under-segmentation, which granularity remedies target, carries a small and operating-point-dependent share.

The wrong-instance class is not a small-mask artefact. Its mean returned IoU against ground truth is 0.006 while its mean oracle IoU is 0.809: the selector is choosing an object, not a fragment, and choosing the wrong one.

C. Wrong-instance is spatial reference failure

Classifying the expressions lexically shows what the wrong-instance cases are made of. 84 % of them contain a spatial word (*left*, *right*, *middle*, *top*, *second from*…) against 67 % of the split overall under the same broad lexical probe; the relation parser used by the operator (§IV) is narrower than that probe and fires on 57.4 % of the split, which is the coverage figure used in Table VII. Typical members are *"left green pot"*, *"person second from left"*, *"right bear"*, *"horse of left"*. The referent is also not buried deep in the ranking: at the unguided operating point its median score-rank is 4.

D. Why the released guidance cannot repair it

HybridGL's guidance modules are not absent, they are *confined*. The released implementation computes a semantic score per proposal, keeps $\mathrm{topk}(\mathrm{score}, k_1{=}3)$, forms a spatial-relationship term and a GEM attention $\times$ position-mask coherence term for those three, and returns

$$\hat{\imath} \;=\; \arg\max_{i \in \mathcal{T}} (1-\alpha)\, r_i \;+\; \alpha\, c_i, \qquad \mathcal{T} = \mathrm{topk}(\mathrm{score}, 3), \tag{1}$$

with r_i the relationship term, c_i the coherence term and $\alpha = 0.6$. The candidate set $\mathcal{T}$ is fixed before any spatial reasoning happens, so no amount of relation or coherence modelling can return a proposal outside it.

TABLE II
CONFINEMENT OF THE RELEASED GUIDANCE ON FULL REFCOCO VAL (10,834 EXPRESSIONS, ORACLE MIOU 0.8221). "CATEGORY SCORE" IS THE SAME SCORER READING THE EXPRESSION WITH ITS SPATIAL VOCABULARY REMOVED.

measurement	value
oracle inside released top-3 (all expressions)	0.639
oracle inside released top-3 (wrong-instance)	**0.302**
mean score-rank of the oracle (wrong-instance)	9.1
oracle inside category-score top-8 (wrong-instance)	**0.701**
mIoU of a perfect chooser inside released top-3	0.5812
mIoU actually achieved	0.4482

Table II quantifies the consequence. Two numbers matter. First, the confinement costs real accuracy: a perfect chooser restricted to the set the released pipeline already computes would reach 0.5812 mIoU against the 0.4482 it achieves — $+13.3$ mIoU is available *inside the released candidate set*. Second, the confinement is what blocks the repair: on the wrong-instance failures the referent is inside $\mathcal{T}$ only 30.2 % of the time, but inside the top-8 of the category score 70.1 % of the time.

The category score is the same frozen text encoder evaluated on the expression after its spatial vocabulary has been removed — *"left green pot"* becomes *"green pot"*. It is a probe for *which objects are candidates at all*, deliberately stripped of the part of the expression that the existing score already handles poorly. That a lexical edit of the prompt produces a better candidate set than the prompt itself is the empirical content of this section.

IV. METHOD: CATEGORY-SET RELATION GUIDANCE

The diagnosis says the binding constraint is the candidate set the released guidance is allowed to act on, and that the category score generates a better set than the full expression. The operator follows directly. It replaces the final selection step and nothing else; it introduces no parameters, performs no training, uses no ground truth, and adds no forward pass — every quantity it consumes is already computed by the pipeline it is applied to.

A. The operator

Given an image x, an expression e, the proposal pool M with per-proposal semantic scores, the released coherence score c_i, and mask geometry:

1) **Category score.** Evaluate the frozen text encoder on e with its spatial vocabulary removed, and score every proposal against it. This is one text encoding per expression, not one per proposal.
2) **Candidate set.** Let $\mathcal{C}$ be the top-k proposals under the category score ($k = 8$ throughout).
3) **Relation parse.** Parse e deterministically for a spatial relation: a horizontal direction (*left*/*right*), a vertical direction (*top*/*bottom*/*above*/*below*), a centrality term (*middle*/*centre*), or an ordinal with an axis (*second from the left*).
4) **Selection.** If a relation was parsed and $|\mathcal{C}| \geq 2$, return the member of $\mathcal{C}$ the relation selects: for an ordinal, the n-th proposal along the relation axis by mask centroid; otherwise the proposal with the highest released coherence score c_i among those in the half of $\mathcal{C}$ lying on the relation's side, split at the median along the relation axis. If no relation was parsed, or $\mathcal{C}$ is a singleton, return the pipeline's own selection unchanged.

Two design choices carry the mechanism and are ablated in §V-E. Ranking is by the *released coherence score*, not by the semantic score and not by geometry: the coherence term already encodes the pipeline's own spatial prior, and §V-E shows that replacing it with the semantic score removes the gain entirely. The side constraint is taken relative to the candidate set's own median rather than the image centre, so the operator asks "which of these candidates is the left one", not "which candidate is in the left half of the image".

B. Ordinal expressions abstain

A deterministic ordinal resolver was implemented and measured: it returns the n-th candidate along the relation axis. On its stratum ($n = 444$ on RefCOCO val) it *loses* 0.153 mIoU relative to the released selection, so the shipped operator abstains on ordinal expressions and the stratum is reported as an open problem rather than silently absorbed. Ordinal reference is 4.1 % of the split.

C. Cost

The operator adds one text encoding per expression, an $O(N \log k)$ partial sort, an $O(k)$ split, and $O(k)$ lookups. It is $O(k)$ per expression after the encoding the pipeline already performs, and it is a pure post-processing step: it can be applied to a stored score matrix without re-running any model, which is how all results below are produced.

V. EXPERIMENTS

A. Protocol

Baseline. HybridGL [4] at its own released operating point, re-implemented from the official code. We report both the *unguided* row (argmax of the semantic score over the full pool, i.e. the "G2L alone" configuration of the paper's ablation) and the *released guidance* row (Eq. 1).

Proposer and scorer. SAM ViT-H with `points_per_side` = 8, `pred_iou_thresh` = 0.7, `stability_score_thresh` = 0.7, `min_mask_region_area` = 800, `crop_n_layers` = 0, giving 48.4 proposals per image on average; CLIP ViT-B/16 with HybridGL's G2L fusion, `masking_block` = 9. All features are computed once and cached; every result below is a re-analysis of one stored pass.

Data. Full validation splits: RefCOCO val (10,834 expressions, 1,500 images), RefCOCO+ val (10,758), RefCOCOg val / UMD (4,896).

Metrics. mIoU (per-expression mean, primary) and oIoU (dataset-level). **Oracle mIoU** is reported for every pool because it is the scorer-independent reference point.

Statistics. Every reported delta is paired over expressions with a 10,000-resample bootstrap (95 % CI) and a sign-flip permutation test (p), with win/loss counts. Any hyper-parameter is selected on a 716-expression split of RefCOCO val (250 references, fixed seed, chosen before the final script was written) and reported on the full split and on the held-out complement.

B. Reproduction

On RefCOCO val our reproduction reaches 0.3518 mIoU / 0.2768 oIoU unguided and 0.4482 / 0.3816 with the released guidance, against the published 40.27 / 31.71 and 49.48 / 41.81. The absolute shortfall is real and §V-H localises it. We report it because it bounds the reading of every absolute number: all comparisons in this paper are internal to one reproduction, and absolute rows are not directly comparable to the published table.

C. Main result

Table III gives the main result. The operator raises the released pipeline on full RefCOCO val from 0.4482 to 0.5029 mIoU and from 0.3816 to 0.4431 oIoU; the delta is $+0.0546$ mIoU (95 % CI $[+0.0493, +0.0601]$, $p < 0.0001$) with 1,721 expressions improved and 764 degraded. Against the unguided argmax the gain is $+0.1510$ mIoU. The oracle-hit rate — the fraction of expressions whose returned proposal is exactly the best pool member — rises from 0.454 to 0.516. Table IV shows the operator is not tuned to a peak: $k \in \{5, 12, 16, 24\}$ gives $+0.0397$ to $+0.0568$, the side-split percentile gives $+0.0455$ to $+0.0546$, and the set source gives $+0.0437$ to $+0.0546$. On the held-out complement of the selection split ($n = 10{,}118$) the delta is $+0.0540\,[+0.0482, +0.0599]$.

D. The operator and the containment snap are complementary, not competing

The most demanding baseline here is a one-line containment snap: replace the selected mask by the smallest pool proposal that covers at least 95 % of it. It is the operator that previously explained away an earlier method of ours, and it is included as a required row.

On mIoU the two are *statistically indistinguishable*: a paired bootstrap on the per-expression difference of the two deltas gives $+0.0005\,[-0.0055, +0.0066]$, $p = 0.87$. We state this plainly rather than reporting the point estimate that favours us. They act on largely different expressions, however — ours is strictly better on 1,324, the snap on 1,323, with 7,314 ties; ours gains where the snap hurts on 45 expressions and the snap gains where ours hurts on 119 — and they separate on everything else that matters:

- **oIoU: 0.4431 vs 0.4072** ($+3.6$ points). The selector moves the dataset-level metric; the snap moves mIoU only.
- **Part expressions** ($n = 443$): $+0.0439$ versus $+0.0345$.
- **GT-area quartiles**: ours is positive in all four ($+0.0329/ + 0.0500/ + 0.0561/ + 0.0795$); the snap's gain is confined to the largest quartile by construction, and it is $+0.0136$ on part expressions at the unguided point.

Because the two fix different error classes — wrong-instance and under-segmentation, §III — they compose almost additively: applying the snap after the selector gives 0.5505 mIoU / 0.4574 oIoU, i.e. $+0.1023$ over the released pipeline. We attribute that row explicitly to the composition rather than to the selector.

E. Controls

Table V isolates the mechanism. Picking the geometric extremum of the candidate set instead of ranking by coherence is *worse than doing nothing* (-0.0549): the operator is not a heuristic preference for extreme positions. Ranking by the semantic score instead of the released coherence score gives -0.0024 (category score) and -0.0117 (full score): the semantic score cannot choose within the set, which is exactly the failure the diagnosis describes. Mirroring the relation — the text says *left*, the resolver takes the rightmost — costs -0.1813, confirming that the relation's semantics rather than any extreme pick carry the gain. Finally, running the same ranking over the whole pool rather than the top-8 category set gives $+0.0484$, and dropping the relation gate gives $+0.0430$.

The decomposition, stated plainly: of the $+0.0546$ total, roughly $+0.043$ comes from allowing the released coherence and position machinery to act on more than its top-3 candidates, $+0.012$ from gating on the parsed relation, and $+0.006$ from building the set with the category score rather than the full expression. The first term is the largest, and it is a structural property of the baseline, not a component we designed.

F. Stratification

By relation type: horizontal ($n = 4{,}749$) $+0.1076$; vertical ($n = 443$) $+0.0735$; middle ($n = 584$) $+0.0820$; ordinal ($n = 444$) 0.0000 (abstains, §IV); no relation ($n = 4{,}614$) 0.0000 by construction. The operator never touches an expression it cannot parse, so the 42.6 % of expressions without a spatial relation are bit-identical to the baseline.

By GT-area quartile the gain is positive throughout ($+0.0329/ + 0.0500/ + 0.0561/ + 0.0795$), in contrast to the containment snap whose benefit is concentrated in large objects. Part expressions gain $+0.0439$.

G. Candidate-set ceilings

For any chooser — including a multimodal model — acting on a given candidate set, the best achievable mIoU is the mean over expressions of the best pool member inside that set. On full RefCOCO val: the released top-3 set admits 0.5812 (the released pipeline achieves 0.4482, i.e. 61 % of it); the operator's top-8 category set admits 0.7303 (the operator achieves 0.5029, 47 % of it); the whole pool admits 0.8221. The operator's candidate set therefore raises the ceiling by $+0.149$ mIoU and captures roughly half of it, and at most $+0.227$ mIoU remains available to a better chooser operating on the same set. This bound is the frame in which the absence of a multimodal chooser baseline should be read (§VI).

TABLE III

FULL REFCOCO VALIDATION SPLIT, 10,834 EXPRESSIONS, RELEASED HYBRIDGL SCORER. ORACLE MIOU 0.8221. Δ IS PAIRED OVER EXPRESSIONS; CI IS A 10,000-RESAMPLE BOOTSTRAP; p IS A SIGN-FLIP PERMUTATION TEST; EVERY $p < 0.0001$. "CHANGED" COUNTS EXPRESSIONS WHOSE RETURNED INDEX DIFFERS FROM THE RELEASED ROW. ALL DELTAS ARE COMPUTED FROM UNROUNDED VALUES, SO A DELTA MAY DIFFER BY ± 0.0001 FROM THE DIFFERENCE OF THE TWO ROUNDED MIOU COLUMNS SHOWN. THE CONTAINMENT SNAP IS A REQUIRED BASELINE AND IS *not* PART OF THE PROPOSED METHOD.

row	mIoU	oIoU	Δ vs released [95 % CI]	Δ vs argmax	changed	win	loss
HybridGL argmax (unguided)	0.3518	0.2768	$-0.0964\,[-0.1030, -0.0898]$	—	4746	1005	2840
HybridGL released guidance	0.4482	0.3816	—	$+0.0964$	0	—	—
+ containment snap (baseline)	0.5024	0.4072	$+0.0541\,[+0.0505, +0.0578]$	$+0.1505$	2995	1826	421
+ ours	**0.5029**	**0.4431**	$+0.0546\,[+0.0493, +0.0601]$	$+0.1510$	2865	1721	764
+ ours + snap (composition)	0.5505	0.4574	$+0.1023\,[+0.0963, +0.1083]$	$+0.1987$	4284	2850	826

TABLE IV

SENSITIVITY OF THE OPERATOR ON FULL REFCOCO VAL. THE SHIPPED CONFIGURATION ($k = 8$, SIDE PERCENTILE 50, CATEGORY SCORE, COHERENCE RANKING) WAS FIXED ON THE 716-EXPRESSION SELECTION SPLIT BEFORE THE FINAL EVALUATION SCRIPT AND IS NOT THE BEST CELL OF ANY SWEEP.

variant	mIoU	oIoU	Δ vs released
shipped ($k = 8$)	**0.5029**	**0.4431**	$+0.0546$
$k = 5$	0.4880	0.4241	$+0.0397$
$k = 12$	0.5050	0.4474	$+0.0568$
$k = 16$	0.5038	0.4470	$+0.0555$
$k = 24$	0.5018	0.4453	$+0.0535$
side percentile 34	0.4938	0.4378	$+0.0455$
side percentile 66	0.5016	0.4389	$+0.0534$
set: stripped text	0.4999	0.4377	$+0.0517$
set: full expression	0.4938	0.4317	$+0.0455$
set: noun probe	0.4919	0.4334	$+0.0437$

TABLE V

CONTROLS ON FULL REFCOCO VAL. EACH REPLACES ONE COMPONENT OF THE OPERATOR AND LEAVES THE REST INTACT.

control	Δ mIoU	reading
geometric extremum of the set	-0.0549	not "take the leftmost"
rank by category score	-0.0024	semantic re-ranking is inert
rank by full semantic score	-0.0117	idem
mirror the relation	-0.1813	the relation semantics are real
whole pool instead of top-8	$+0.0484$	category set: $+0.006$
coherence ranking, no gate	$+0.0430$	side gate: $+0.012$

H. Reproduction gap, localised

The published ablation of HybridGL gives G2L alone 31.71 oIoU / 40.27 mIoU and G2L with all three guidance modules 41.81 / 49.48 on RefCOCO val, i.e. a guidance increment of $+10.10$ oIoU / $+9.21$ mIoU. Our reproduction's guidance increment is $+10.48$ / $+9.64$ — within 0.4 points. The entire shortfall therefore lies in the frozen feature/score path, not in the guidance we re-implemented:

dataset	base (G2L) gap	guidance gap
RefCOCO val	-4.03 oIoU / -5.09 mIoU	$+0.38$ / $+0.43$
RefCOCOg val	-2.32 / -2.16	$+1.29$ / $+0.99$

This matters twice over. It establishes that the guidance modules — the component our operator interacts with — are faithfully reproduced, and it scopes the absolute numbers: a better base feature path would raise both rows together and is orthogonal to this paper's claim.

TABLE VI

PROPOSAL-POOL DENSITY ON THE IDENTICAL 716 SELECTION-SPLIT EXPRESSIONS. DOUBLING THE POOL BUYS $+1.0$ MIOU OF ORACLE WHILE DEGRADING THE RELEASED PIPELINE BY 3.2 MIOU, AND LEAVES BOTH THE DIAGNOSIS AND THE OPERATOR INTACT.

	`pps`= 8	`pps`= 16
proposals / image	48.4	101.7
oracle mIoU	0.8285	0.8388
released mIoU	0.4700	0.4383
wrong-instance gap share	0.741	0.720
prevalence	0.344	0.363
under-segmentation share	0.205	0.226
ours mIoU	0.5329	0.4968
Δ vs released	$+0.0629$	$+0.0585$
Δ on spatial stratum	$+0.1041$	$+0.0968$

I. Pool density does not explain the diagnosis

Every other measurement in this paper uses one pool. If the diagnosis were an artefact of a sparse pool, a denser one would dissolve it. Table VI re-runs the entire released pipeline with SAM at `points_per_side` $= 16$ on the identical 716 expressions ($+110\,\%$ proposals, 48.4 $\rightarrow$ 101.7 per image).

Doubling the pool buys only $+1.0$ mIoU of oracle, which is the diagnosis in one number: the pool was not the constraint. Worse for the pool-centric view, the released pipeline *degrades* by 3.2 mIoU, because a denser pool adds distractors and the selector must now choose among more of them. The wrong-instance share of the gap is stable ($0.741 \rightarrow 0.720$), and the operator still works ($+0.0629 \rightarrow +0.0585$ overall, $+0.1041 \rightarrow +0.0968$ on spatial expressions). A better pool does not repair selection failure; it concentrates it.

J. Cross-dataset behaviour, and the boundary of the repair

Table VII and Fig. 2 report the operator on all three splits. Three readings.

The diagnosis transfers. Wrong-instance carries 77.4 %, 86.4 % and 80.1 % of the recoverable IoU on RefCOCO, RefCOCOg and RefCOCO+ respectively, all at the released operating point; on RefCOCO it carries 82.2 % at the unguided point as well, and 72.0–74.1 % at both pool densities tested. The central claim therefore holds on all three standard benchmarks, at two operating points on RefCOCO, and at two pool densities.

The repair does not transfer uniformly, and the reason is measurable. RefCOCO+ contains a spatial word in only

TABLE VII
CROSS-DATASET BEHAVIOUR OF THE OPERATOR AT THE RELEASED HYBRIDGL OPERATING POINT. THE THREE SPLITS DIFFER SHARPLY, AND FOR CONSTRUCTION REASONS, IN HOW OFTEN AN EXPRESSION CARRIES A SPATIAL RELATION: REFCOCO+ WAS COLLECTED WITH LOCATION WORDS EXCLUDED. "Δ FIRED" IS THE DELTA RESTRICTED TO EXPRESSIONS WHOSE RETURNED INDEX THE OPERATOR CHANGED. WRONG-INSTANCE SHARE IS THE FRACTION OF THE ORACLE GAP MASS THAT CLASS CARRIES.

dataset	n	spatial %	fires %	released mIoU	ours mIoU	Δ all	Δ spatial	Δ fired	wrong-inst. share
RefCOCO	10,834	57.4	26.4	0.4482	0.5029	+0.0546	+0.0951	+0.2065	0.774
RefCOCOg	4,896	17.7	8.3	0.5008	0.4974	−0.0034	−0.0191	−0.0406	0.864
RefCOCO+	10,758	**1.6**	0.8	0.4023	0.4035	+0.0012	+0.0791	+0.1591	0.801

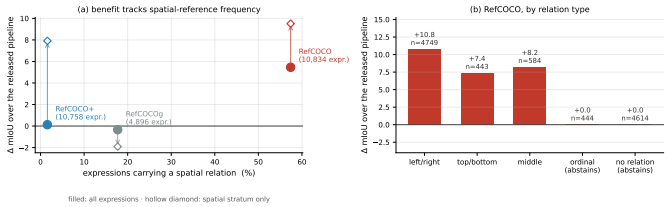


Fig. 2. (a) The operator's benefit tracks how often expressions carry a spatial relation. Filled markers: all expressions; hollow diamonds: the spatial stratum only. The RefCOCO+ outcome was predicted before the run, from the dataset's construction. (b) Deltas by relation type on RefCOCO. Ordinal and relation-free expressions are unchanged by construction.

1.6 % of its expressions, by construction; the operator fires on 0.8 % of them and is neutral overall (+0.0012). RefCOCOg occupies the middle (17.7 %) and is also neutral (−0.0034). RefCOCO, at 57.4 %, gains +0.0546.

The neutral RefCOCO+ result is a low-information null, not a failure, and we test it that way. A neutral overall number would be uninformative if the operator were simply broken there. Restricted to the 1.6 % of RefCOCO+ expressions that do carry a spatial relation, the operator gains +0.0791 overall, with +0.0558 / +0.0944 / +0.1311 on the horizontal ($n = 8$) / vertical ($n = 105$) / middle ($n = 23$) sub-strata. The two substantive sub-strata have the same sign and comparable magnitude as on RefCOCO. The horizontal sub-stratum has $n = 8$ and we do not draw conclusions from it.

RefCOCOg is the genuine exception: neutral overall *and* slightly negative on its own spatial stratum (horizontal +0.0149, $n = 563$; vertical −0.1266, $n = 176$; middle −0.0342, $n = 78$). RefCOCOg's expressions are long and predominantly *object-relative* ("the man on the left in the blue shirt", "the bird above the branch"), for which an image-frame coordinate resolver is the wrong instrument. We report this as a boundary with a hypothesis rather than tuning the operator until the hypothesis fits.

K. Qualitative behaviour

Fig. 3 shows expressions on which the operator changes the answer. The pattern is uniform and matches the diagnosis: the released pipeline selects a real object of the correct category — another person, another sheep, another horse, another car — and the operator selects the referred one. This is the failure mode that dominates the oracle gap table, made visible.

VI. LIMITATIONS AND THREATS TO VALIDITY

We state the limits of this work in the order in which they bound its claims.

No multimodal chooser baseline. The most obvious alternative to a hand-written relation resolver is to hand the candidate set to an off-the-shelf multimodal model and ask it to pick. We did not run that baseline. The local reproduction of GLOD-RIS, whose pipeline depends on such a model, broke in production (aborted at 223 of 10,834 expressions with a malformed structured response), and no local vision–language weights are available in this environment. We bound the question instead (§V): any chooser acting on the operator's candidate set could add at most +0.227 mIoU, and the internal alternatives we did test — ranking by the semantic score, and picking the geometric extremum — both lose to the coherence ranking. This is the largest open flank of the paper and we flag it as such rather than reporting the ceiling as if it were the comparison.

The mIoU gain is not distinguishable from a one-line baseline. On full RefCOCO val the operator's mIoU delta is statistically indistinguishable from a containment snap (+0.0005, $p = 0.87$). The selector separates on oIoU (+3.6 points), on part expressions, and on coverage of the GT-area quartiles, and the two compose almost additively. A reader who cares only about mIoU on this benchmark should read the two rows as tied.

The repair does not transfer uniformly. It is neutral on RefCOCOg (−0.0034) and RefCOCO+ (+0.0012). On RefCOCO+ the null is expected — 1.6 % of expressions are spatial — and the residual spatial stratum is positive. On RefCOCOg the operator is negative on its own target stratum, which we attribute to object-relative spatial language. A method that helps on one of three benchmarks is a scoped contribution, and we scope it accordingly.

One proposal generator and one vision–language backbone. All results use SAM ViT-H and CLIP ViT-B/16. Pool *density* was varied (`points_per_side` 8 → 16, Table VI) and both the diagnosis and the operator survived, but a different proposal generator or a different vision–language backbone was not tested. The taxonomy's robustness additionally rests on three scorers spanning roughly 10 mIoU on RefCOCO, but those three share the same frozen features.

Reproduction gap. Absolute rows are not comparable to the published HybridGL table: our reproduction is 4.03 oIoU / 5.09 mIoU below the published G2L row on RefCOCO val and 2.32 / 2.16 on RefCOCOg, while the guidance increment matches within 0.4 points (§V-H). Since all head-to-head

comparisons are internal to one reproduction, the gap does not affect them; it does mean the absolute numbers should not be quoted as state-of-the-art figures.

Ordinal reference is unsolved. Ordinal expressions are 4.1 % of RefCOCO val; the deterministic ordinal resolver we implemented lost 0.153 mIoU on that stratum, so the operator abstains and the stratum is untouched. Ordinal and depth reference remain open.

The spatial vocabulary is a fixed list. The parser matches a hand-written vocabulary of English spatial terms rather than learning it. Tier ablations at the unguided operating point show the gain is dominated by left/right alone ($+0.1064$) and increases monotonically as middle/centre ($+0.1171$), ordinals ($+0.1292$) and the full list ($+0.1331$) are added, but the list is still a design commitment, and languages or phrasings outside it are not handled.

The evaluation is on RefCOCO-family benchmarks only. No generalized (no-target) or multi-target benchmark was run; the taxonomy and the operator say nothing about those settings.

VII. Conclusion

Training-free referring image segmentation has been optimised under the assumption that the proposal pool is the limiting factor. We tested that assumption by decomposing the oracle gap — the IoU that the pool makes available minus the IoU the selector returns — into attributable error classes, on the full RefCOCO, RefCOCO+ and RefCOCOg validation splits, at two operating points of the released HybridGL pipeline on RefCOCO, and at two proposal-pool densities.

The pool is not the bottleneck. *Wrong-instance* selection, in which the pipeline returns a different object of the correct category, carries 77–86 % of all recoverable IoU on all three benchmarks, while under-segmentation — the granularity class that mask-merging and mask-refinement remedies address — carries 5–23 %. Doubling the proposal density confirms the reading from the other direction: it buys $+1.0$ mIoU of oracle and costs the released pipeline 3.2 mIoU, because more proposals means more distractors to select among.

The mechanism is structural rather than incidental. The released guidance computes a spatial-relationship term and a coherence term, but evaluates them only on the three highest-scoring proposals under the full expression, so its answer cannot leave that set; on wrong-instance failures the referent is outside it 69.8 % of the time. A perfect chooser restricted to the set the pipeline already computes would gain $+13.3$ mIoU, and a lexical edit of the prompt — removing the spatial vocabulary, which turns the expression into a category probe — moves the referent into the top-8 for 70.1 % of those failures.

Acting on that, a training-free operator that enlarges the candidate set and applies the pipeline's own coherence score subject to the parsed relation raises the released pipeline to 0.5029 mIoU and 0.4431 oIoU on full RefCOCO val ($+0.0546$ mIoU, 95 % CI $[+0.0493, +0.0601]$), with the gain concentrated exactly on spatial expressions and zero on the 42.6 % of expressions it cannot parse.

We have been deliberately explicit about what this does not establish. The mIoU gain is statistically indistinguishable from a one-line containment snap; the operator is neutral on RefCOCOg and RefCOCO+; no multimodal chooser baseline was run and that remains the largest open flank, bounded at $+0.227$ mIoU by the candidate-set ceiling; and absolute rows are not comparable to the published table because our reproduction of the baseline is 5.09 mIoU below it on the base row, a gap we localise to the frozen feature path.

The durable result is the diagnosis. It reframes what a training-free RIS method should be measured against: not the oracle utilisation of its pool, but the share of the pool's recoverable IoU that its selector can actually reach. On the evidence here, that share is around half, and the missing half is overwhelmingly a matter of telling one instance from another.

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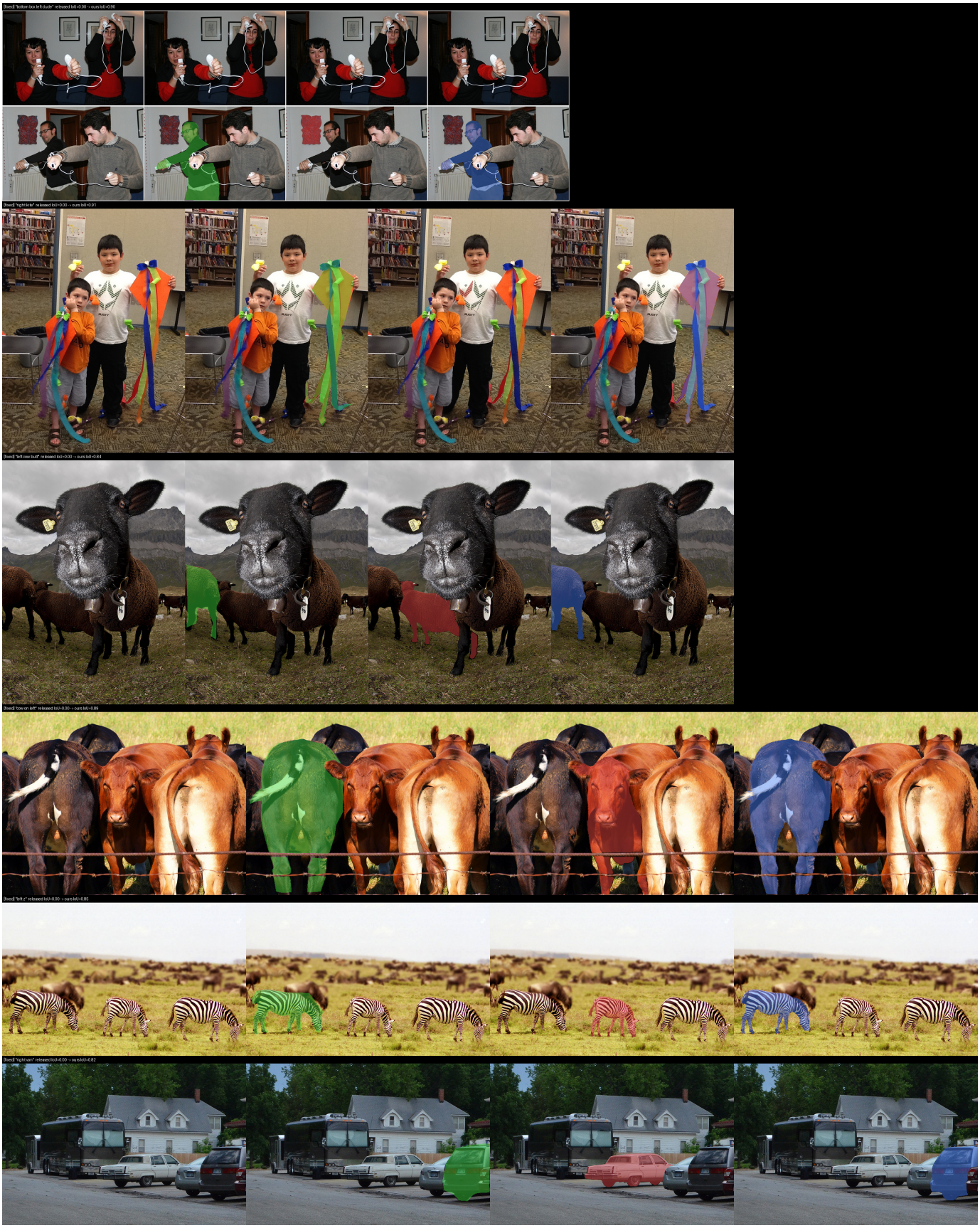


Fig. 3. Expressions the operator repairs. Columns: input; ground truth (green); the released HybridGL selection (red); ours (blue). Rows are ordered by reference id and are a deterministic sample, not a curated selection. In each case the released pipeline returns a plausible instance of the right category and the operator returns the referred one.